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IUT Medical Centre Requirement Elicitation

Emergency, Diagnostic, and Referral Services

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Team Members

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Scope Overview

Emergency, Diagnostic & Referral Services

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Assigned Scope

- Emergency
- Diagnosis
- Referral Services

this includes campus medical centre emergency intake, triage and ambulance dispatch as well

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Stakeholder Analysis

Roles, Interests & Influence

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Primary Stakeholders

- Patients : Student, Faculty & Stuff
- Campus Nurses
- Campus Doctors

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Secondary Stakeholders

→ Ambulance Service Providers

- Vehicle Providers
- Ambulance Drivers
- Maintenance Personnel
- Backup Transport Support

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External Stakeholders

→ Partner hospitals, clinics and diagnostic centres

- Accept referred patients
- Provide specialized treatment
- Share diagnostic services
- Reimburse medical fees

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Targeted Referral Hospitals

- Holy Family Red Crescent Medical College Hospital
- Uttara Women's Medical College Hospital
- International Medical College Hospital

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Methodologies & Key Findings

Insights gathered through field investigation

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Methodologies

- On-site interviews
- Scenario walkthroughs
- Student survey
- Document Reviews

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Finding 1 : Communication Gap

Observed Issues

- Slow paper-based referrals
- Low hotline awareness

Impact

- Slower ambulance dispatch
- Delayed emergency assistance

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Finding 2 : Key Bottleneck

Observed Issues

- Mandatory profile completion
- Manual patient registration

Impact

- Delayed emergency care
- Increased waiting time

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Finding 3 : Core User Demand

Observed Issues

- Digital patient portal
- Referral tracking

Impact

- Faster access to services
- Greater transparency

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User Requirements

Role-Based Needs and Expectations

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Nurses : Emergency intake and Triage

- Emergency bypass registration
- Priority-based triage assignment
- Real-time patient queue management
- Quick patient lookup
- Instant referral initiation

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Doctors : Diagnosis and treatment

- Review diagnostic results
- Create digital referrals
- Approve treatment decisions
- Access patient medical history
- Referral acknowledgment and tracking

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Drivers : Ambulance Operations

- Instant dispatch notifications
- Real-time status updates
- Route and destination details
- ETA tracking
- Emergency assignment acceptance

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Technicians : Diagnostic Services

- Upload test results digitally
- Update test processing status
- Manage diagnostic requests
- Notify result availability
- Access patient test records

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Admin : System administration and reporting

- Manage user accounts
- Update provisional records
- Monitor system activity
- Generate utilization reports
- Access operational analytics

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Admin : System administration and reporting

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Functional Requirements

Core System Features and Capabilities

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Intake & Triage

- Offline data capture with auto-sync
- 3-level priority classification
- Real-time queue management

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Referrals

- Unique referral ID generation
- Digital referral documents
- Electronic signature support

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Ambulance

- Automated dispatch alerts
- Driver status tracking
- Live location updates

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Diagnostics

- Digital result uploads
- Automated notifications
- Patient portal access

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Security & Operations

- Role-based access control
- Complete audit logs
- Persistent emergency hotline

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Non-Functional Requirements

Quality Attributes and System Constraints

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Performance

- <2 sec page loads
- <5 min result notifications
- <10 sec report generation

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Availability

- 99%+ uptime
- 50 concurrent users
- 200 daily events

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Reliability

- Hourly backups
- Nightly full backups
- RPO $\leq$ 1 hr, RTO $\leq$ 2 hr

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Security

- TLS 1.2+ encryption
- AES-256 data protection
- Multi-Factor Authentication

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Portability

- Responsive web interface
- Android & iOS support
- Offline functionality

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Assumptions & Challenges

Key project dependencies and implementation risks

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Assumptions

- Trained personnel are available
- Campus Wi-Fi/Intranet is reliable
- SMS and email services remain operational

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Technical Challenges

- Offline synchronization conflicts
- Digitizing legacy paper records
- Data migration and validation

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Operational Challenges

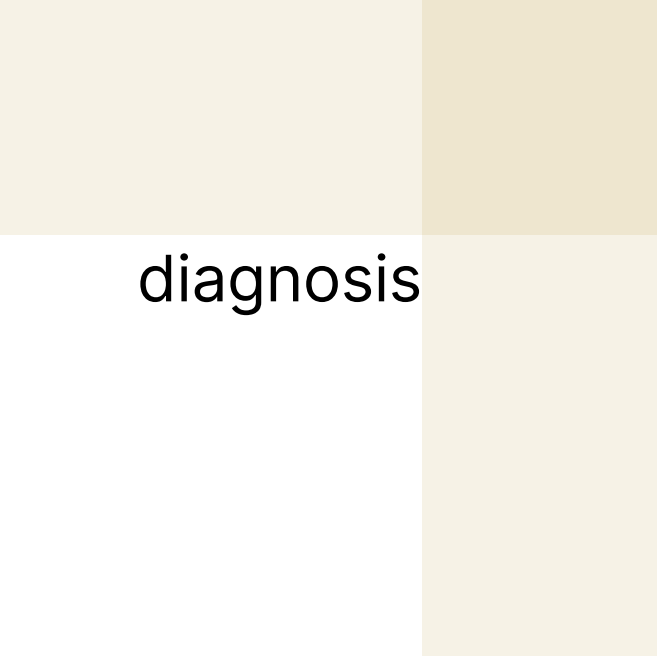
- Privacy and consent compliance
- Staff resistance to digital adoption
- Limited stakeholder availability

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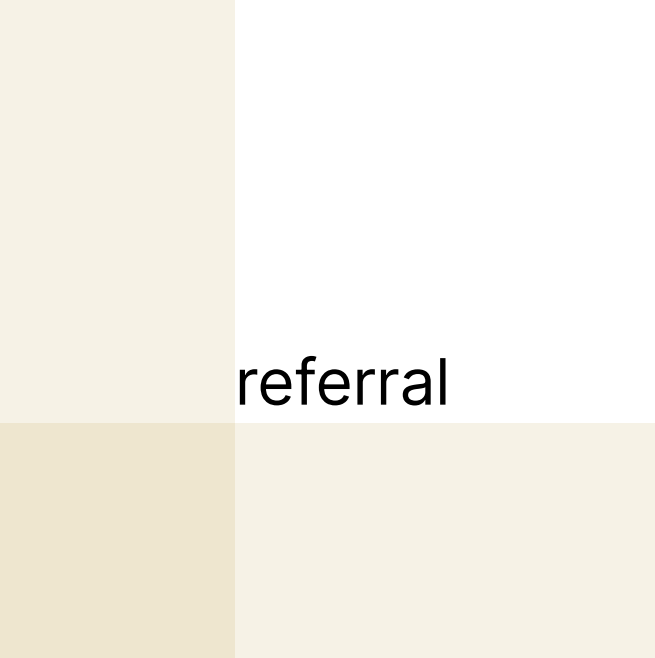


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That's All For Today



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